Acknowledgements

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We would like to thank the international Bebras community for allowing us to use the tasks they have developed. Bebras is a collective effort of many countries.

We would like to thank ACM India's CSpathshala initiative volunteers for localizing and reviewing the tasks and reaching out to schools. We would also like to thank the teachers who have put in the effort for their students to participate in the challenge.

The online version of the competition was implemented on cuttle.org. We would like to thank the following for their excellent collaboration: Eljakim Schrijvers, Justina Oostendorp, Alieke Stijf, Kyra Willekes, Dave Oostendorp: cuttle.org, Netherlands

The 2026 Bebras India competition was organized by ACM India's CSpathshala initiative with support from Tata Consultancy Services. Raspberry Pi Foundation India (https://www.raspberrypi.org/) and Educational Initiatives (https://https://ei.study/) provided several tasks from India.

This booklet of tasks was created on August 9, 2026 using the LaTeX typesetting system developed by Christian Datzko and the Bebras toolchain developed by Jean Philippe Pellet. We extend our immense gratitude to the entire Swiss Bebras team for generously sharing this system with us. In particular, we would like to thank Jean Philippe Pellet and Susanne Falkenberg-Thut for their continuous guidance and support in making the system fully operational.

Bebras India 2026

Bebras (www.bebras.org) is an international student Computational Thinking Challenge organised in over 95 countries and designed to get students from all over the world excited about computing. The challenge is a great way to learn about computational thinking and problem solving skills.

The Bebras challenges are made of a set of short problems called Bebras tasks. The tasks are fun, engaging and based on problems that Computer Scientists enjoy solving. The tasks require logical thinking and can be solved without prior knowledge of computational thinking. The aim is to solve as many as you can in the allotted time.

Bebras India Challenge (www.bebras.in) is organized by ACM India's CSpathshala initiative (www.cspathshala.org). ACM India/CSpathshala's goal is to make it possible so that every child in India learns Computer Science.

The 2026 Bebras India Challenge was held from 16th to 30th November 2026. The 2026 Bebras India Computational Thinking Challenge was offered for age groups 8-19 in 12 languages: English, Marathi, Gujarati, Hindi, Bangla, Bodo, Punjabi, Kannada, Odia, Tamil, Telugu and Assamese and was conducted for free.

Challenge Site: bebras.cspathshala.org
Queries? Contact: challenge@bebrasindia.org

Challenge Groups

Bebras challenge was conducted in six groups:

Group	Ages	Standards	Tasks
Aryabhata	8–10 years	Std III–IV	12
Brahmagupta	10–12 years	Std V–VI	15
Bhaskara	12–14 years	Std VII–VIII	15
Mahavira	14–16 years	Std IX–X	15
Ramanujam	16–18 years	Std XI–XII	15
Pingala	19 years	First Year	15

Scoring

Each task is sorted in a category based on the difficulty levels: A, B or C. Students started with some points and could gain or lose points as follows:

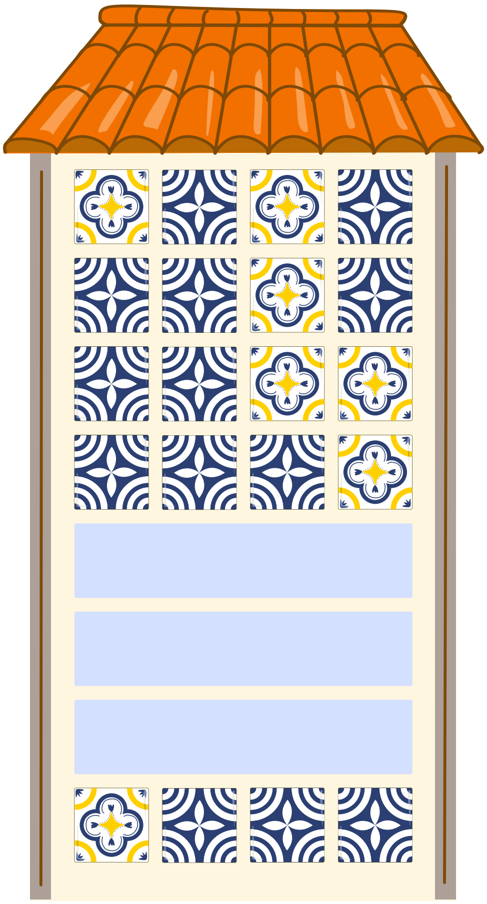
Difficulty	Correct Answer	Incorrect Answer	Unanswered
A category questions	+6 points	0 points	0 points
B category questions	+9 points	-2 points	0 points
C category questions	+12 points	-4 points	0 points

On the following pages you will find the tasks used in 2026 Bebras India Challenge. After each question there is an answer, an explanation of how the answer could be obtained plus a section on how the tasks are related to Computational Thinking.

Contents

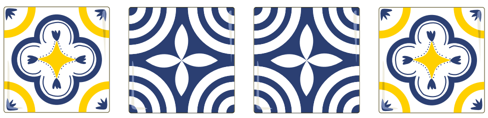
1 Ajulejos

A house wall is to be covered with azulejos (Portuguese tiles). The pattern consists of yellow tiles  and blue tiles . These are arranged in such a way that exactly one tile must change from row to row.

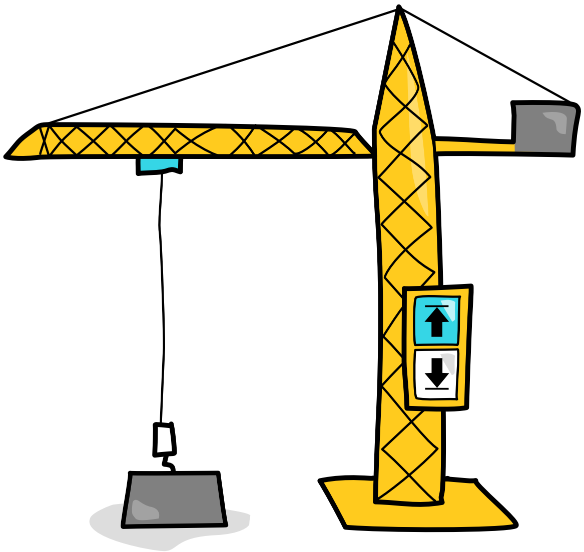


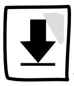
Three rows are missing from the pattern. Which option shows the correct second missing row?

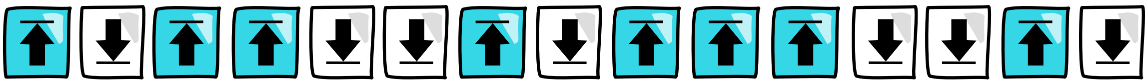
A.  B. 

C.  D. 

2 Toy Crane



This toy crane has two buttons that move the box either up  or down . At the start of the day, the box is at the bottom. The box moves only when it is possible to do so. For example, if the box is already at the top, pressing  again does not move it any higher. Amit pressed the buttons randomly:



How many times did the box move upwards?

A) 4

B) 5

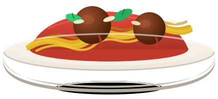
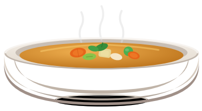
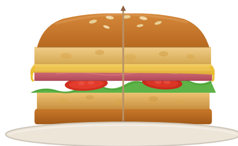
C) 8

D) 10

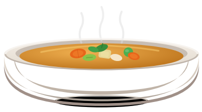
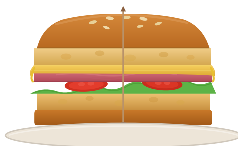
3 What's for Lunch

Bebrasville school serves lunch every day. The kitchen follows a rotating menu that repeats every 4 days in the same order:

The menu started on day 1 with pasta and continued with rice, soup and sandwich. On day 5 they serve pasta again and so on.

Day 1	**Day 2**	**Day 3**	**Day 4**
			
pasta	rice	soup	sandwich

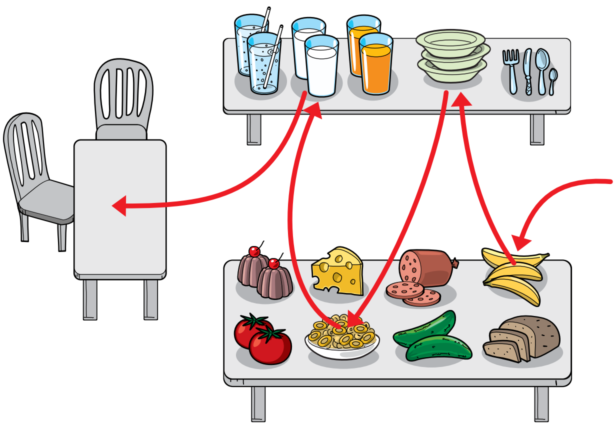
Which dish will be served on day 11?

			
A)	B)	C)	D)

4 Breakfast

These are two breakfast tables.

Marko walked to get things as the red path shows.



Which of the following statements is incorrect?

A) Marko got milk last.

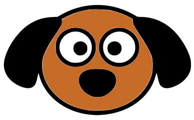
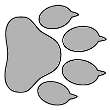
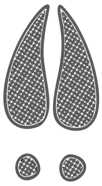
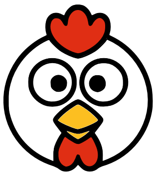
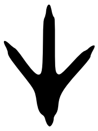
B) After the bowl, Marko got chakli.

C) Marko did not get tomato.

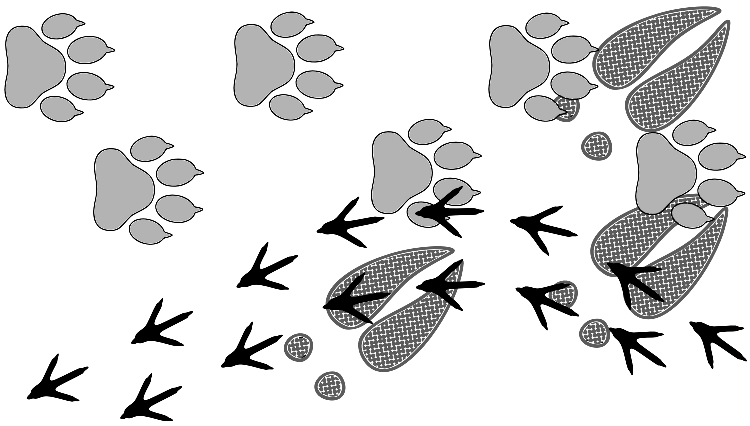
D) First, Marko got a bowl.

5 The Mysterious Footprints

Bunny was walking in the forest after a light snowfall. He noticed some animal footprints in the snow. The animals whose footprints he saw are listed below.

Animal	**Footprints**
 dog	
 deer	
 chicken	

The footprints he saw looked like this:



In what order did the animals walk through the forest?

A) Dog → Deer → Chicken

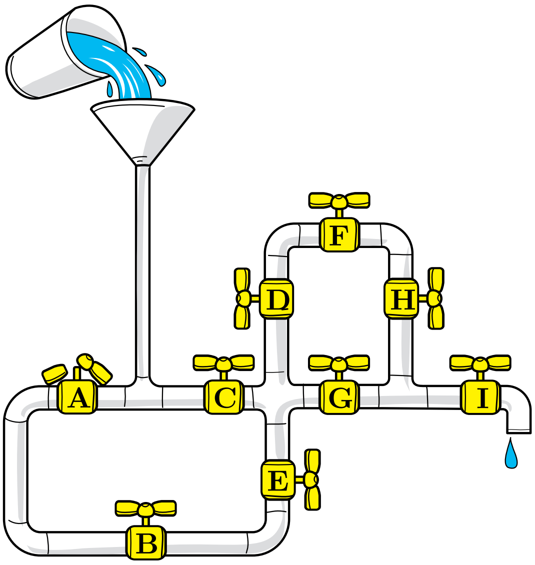
B) Deer → Dog → Chicken

C) Chicken → Dog → Deer

D) Deer → Chicken → Dog

6 Stop the Water

Bunty is playing with pipes and pouring water into them. Along the pipes, there are taps labelled A to I that can be opened and closed to control the flow of water. But tap A is broken and cannot be closed.



Which taps should be closed to prevent water from reaching tap E, using the fewest possible taps?

A) Taps A and C

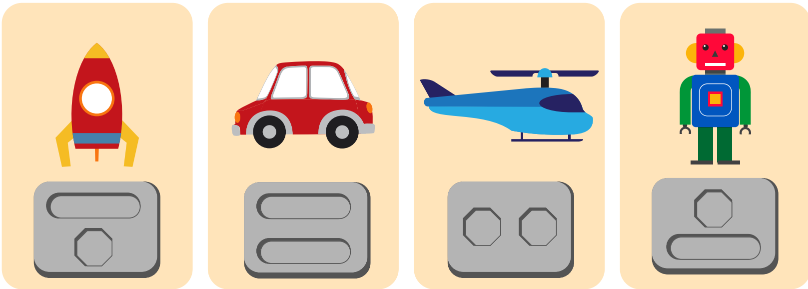
B) Taps C and F

C) Taps B, C and F

D) Taps B and C

7 Toys and Batteries

Milu has some toys that need batteries to work. The shapes of batteries needed for each toy are shown in the picture below.



Milus mother found some batteries at home that can be used in these toys. All of these batteries shown in the picture below can only be used once.

Milu wants to power as many toys as possible at the same time

What is the greatest number of toys that can be powered at the same time using these batteries?

A) 1
B) 2
C) 3
D) 4

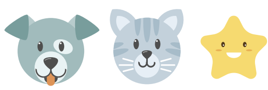
8 Amelia's Message

Amelia sends messages to her friends using animal stickers  .

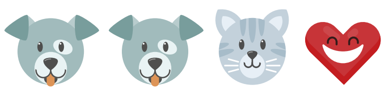
At the end of each message, Amelia adds 1 or more special stickers as follows:

- If the message contains exactly two animals, she adds a star:
- If the message contains exactly three animals, Amelia adds a heart: 
- If all the animals in the message are all different from each other, Amelia adds a sun:

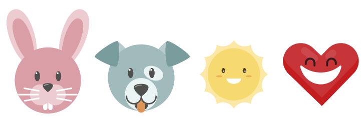
Which of the following messages could Amelia have sent??



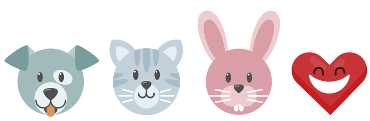
A)



B)



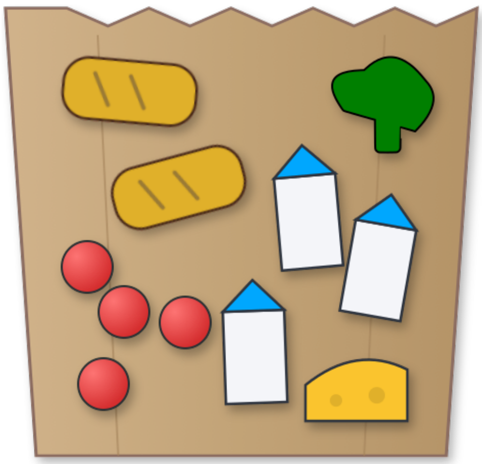
C)



D)

9 Shopping

Lalit went shopping.



BALAJI SUPERMART

ITEM CODE	PRICE
123484929354............	₹10.00
123484920354............	₹20.00
123484921354............	₹30.00
123484922354............	₹40.00
123484929354............	₹10.00
123484921354............	₹30.00
123484922354............	₹40.00
123484929354............	₹10.00
123484923354............	₹50.00
123484929354............	₹10.00
123484922354............	₹40.00

TOTAL ₹290.00

ITEMS COUNT: 11

THANK YOU FOR SHOPPING!
Please come again.

What is the price of one bread *?*

A) Rs.10

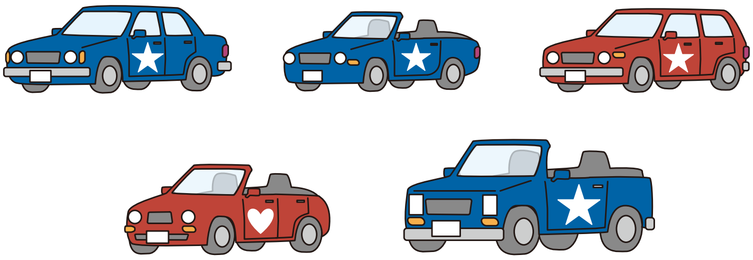
B) Rs.20

C) Rs.30

D) Rs.40

10 Car Inspection Robot

A factory makes five different types of car and each one has a different shape.



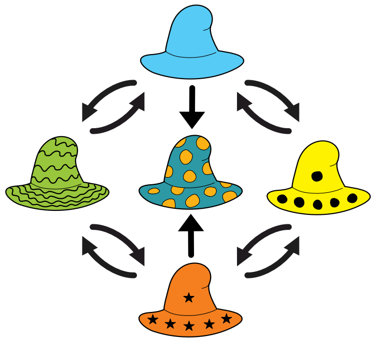
Dr. Beaver wants a robot to identify each of these five car types. However, the robot is old and cannot recognize them by shape. It can only recognize three characteristics.

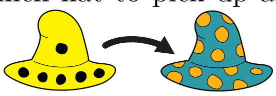
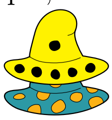
Which combination of three characteristics will allow the robot to tell all five cars apart?

A) Car color, door logo, and number of doors
B) Car color, door logo, and whether the roof is open
C) Door logo, number of doors, and whether the roof is open
D) Car color, whether the roof is open, and headlight shape
E) Door logo, number of doors, and headlight shape

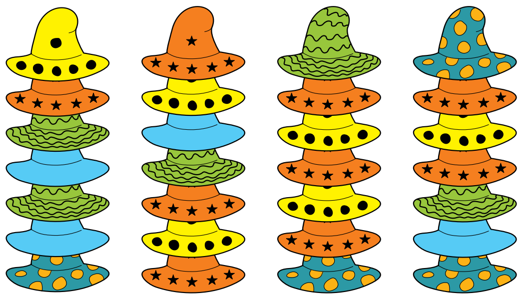
11 Tower of Hats

A hat shop owner likes to create attractive displays by arranging hats into towers. However, he does not want to stack the hats randomly. Instead, he asked an engineer to design an elegant arrangement scheme:



Each arrow shows which hat to pick up and place directly on top of the hat already there. For example, the scheme  shows that the yellow (left) hat is stacked on the right one: 

The shop owner intended to follow the scheme and built four towers of hats. However, he made a mistake in one of the towers. Which tower does not follow the scheme?



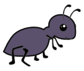
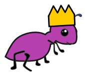
A) B) C) D)

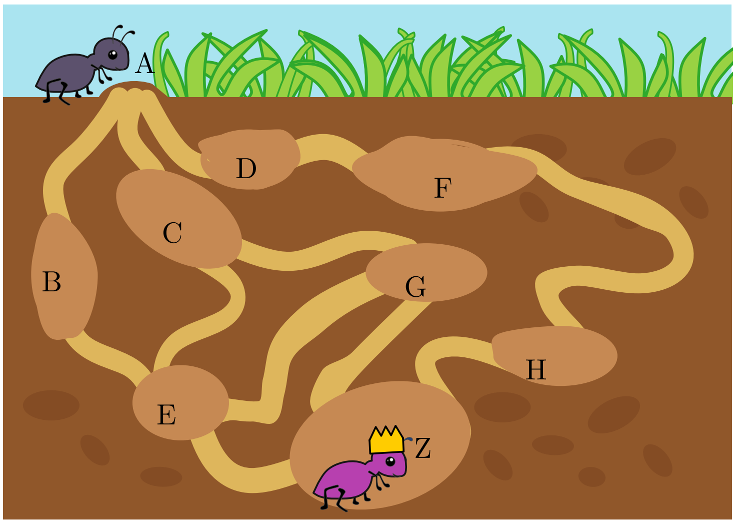
12 The Safe Tunnel

The ants use a complex network of underground tunnels to move around.

Recently, a very sneaky Spider has invaded the colony. She spins an **invisible web** to trap the passing ants. Nobody knows exactly where the Spider is hiding today, but the ants have noticed a pattern in her behavior:

The Spider is too afraid to go to the Entrance (A) or the Queens room (Z), so those are always safe. However, she likes to place her webs **only** in busy rooms, the spots **where 3 or more tunnels meet**.

A Scout Ant  needs to travel from the Entrance (A) to the Queen  (Z) as shown in the map.



Based on the tunnel map, Select the path that guarantees the ant will arrive safely, no matter which of the busy rooms the Spider chose today.

A) A → C → G → Z

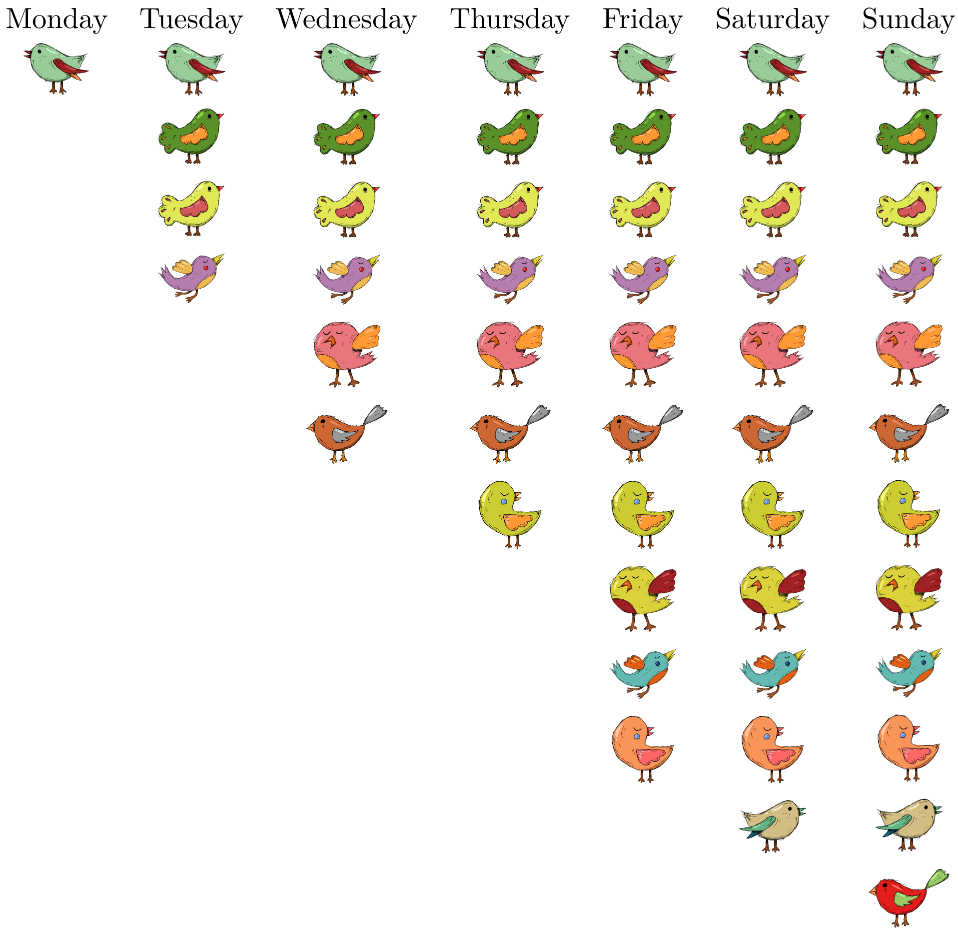
B) A → D → F → H → Z

C) A → B → E → Z

D) A → C → E → G → Z

13 Bird Watching

A school environment club spent one week bird watching. They kept a list which they updated every time they saw a type of bird new to the club. This is what the list looked like after each day:



Some members of the club could only participate on four consecutive days.

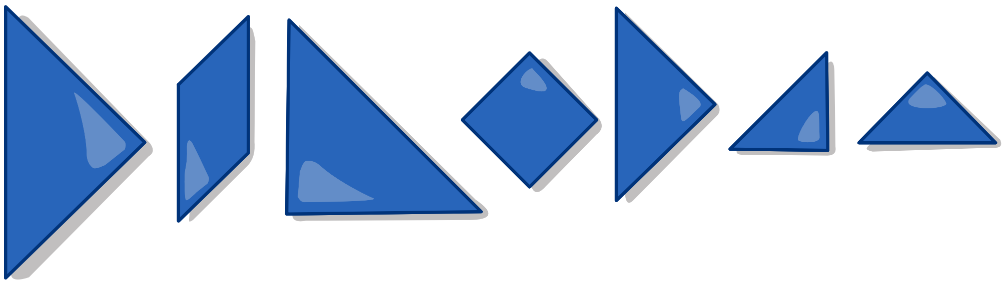
- Anita was there from Tuesday to Friday.
- Badri was there from Wednesday to Saturday.
- Chetan was there from Monday to Thursday.
- Diya was there from Thursday to Sunday.

Which of these four members was present for the most sightings of a type of bird new to the club?

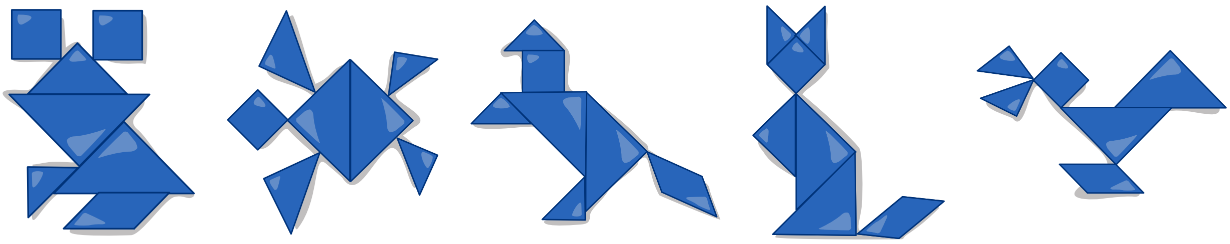
A) Anita
B) Badri
C) Chetan
D) Diya

14 Beaver Puzzle

Two beavers, Geeta and Palak, are playing a puzzle together. The puzzle uses seven simple pieces. They have a picture sheet that shows different animals. One by one, they try to put together the animals using the seven pieces. The pieces can be rotated but cannot cover each other.



Which animal can they form using all seven pieces exactly once?



A) B) C) D) E)

15 Laptops

Four pupils attend an computer science course. At the end of class, they always store their laptops in two boxes. They stack the laptops carefully so that the next day, each pupil can grab their own directly off the top without having to move or rearrange any of the others.

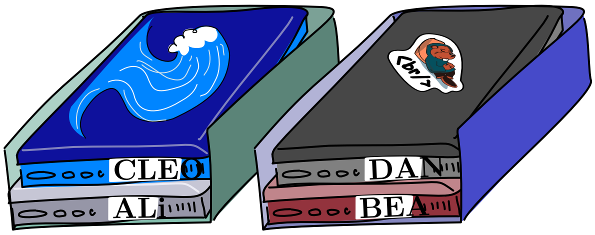
Today, they left the room in this order:

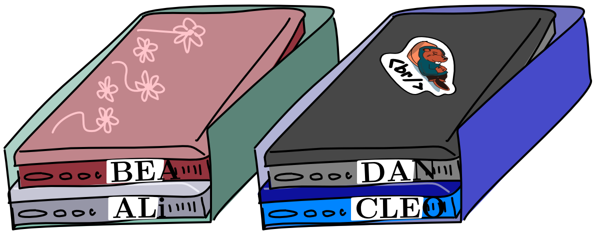
First Ali, then Bea, then Cleo, and finally Dan.

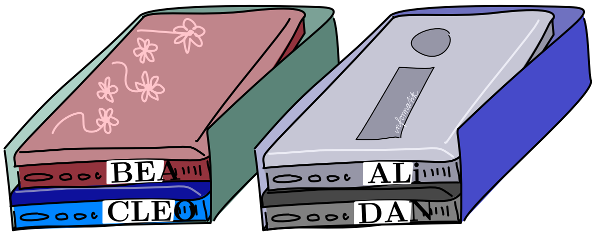
Based on their timetables, they already know they will arrive tomorrow in this order:

First Bea, then Ali, then Dan, and finally Cleo.

How did they arrange their laptops when they left the room?

A) 

B) 

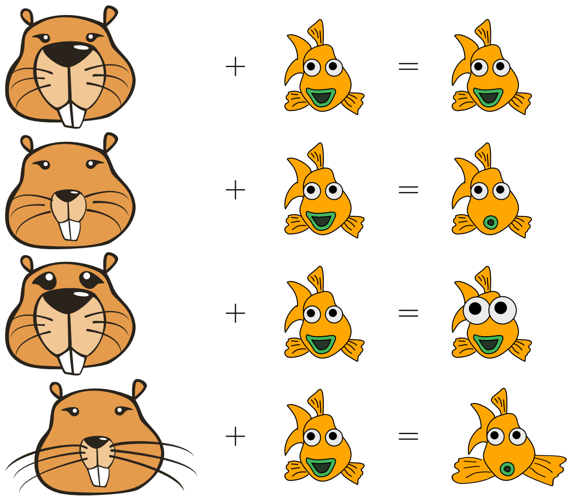
C) 

D) 

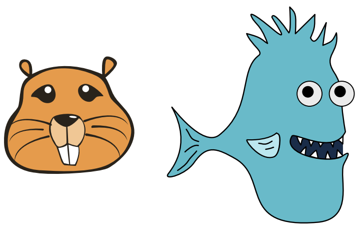
16 Beaverator

The Beaverator is a machine that uses two pictures as input: one of a beavers face and one of a fish. The machine may change the fish picture. Then it outputs the resulting picture.

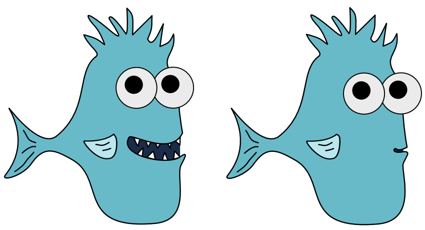
The Beaverator always follows the same simple rules to decide how to change the fish picture. The way a specific part of the beavers face looks for example its eyes tells the Beaverator to change a specific part of the fish in a similar way. See four examples of the Beaverators operation:



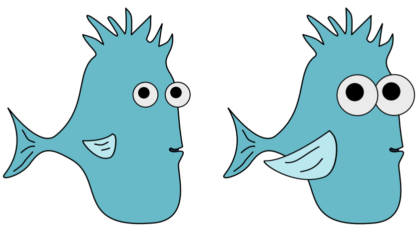
What picture will the Beaverator produce for these two inputs?



A) B)

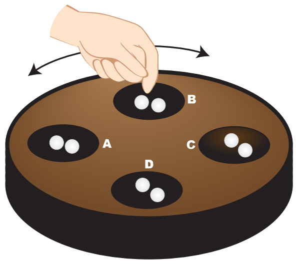


C) D)



17 Pallanguzhi

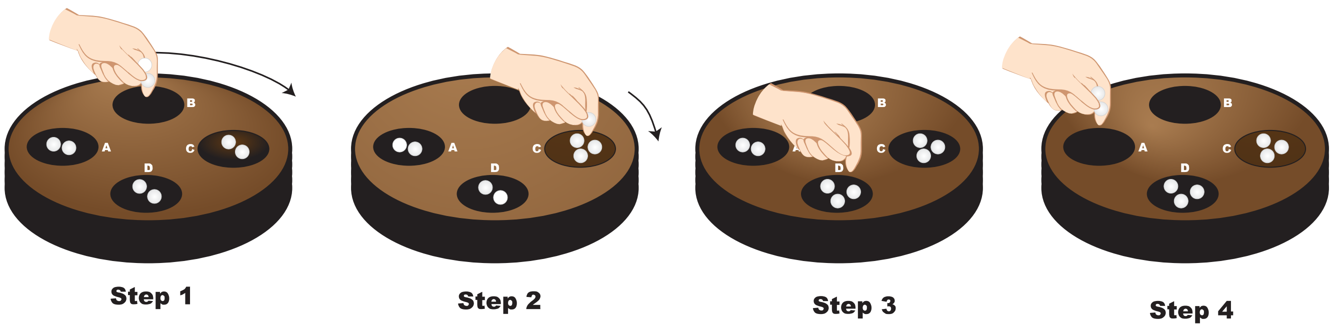
Chikki the beaver, loves playing ńPallanguzhi,ż a traditional board game from India. The wooden board has 4 pits A, B, C, and D arranged in a circle. Each pit starts with 2 beads.



Chikki plays using these rules: 1: She chooses a pit. Then she picks up all the beads from it. 2: If the number of beads she picked up is even, she moves her hand around the board clockwise dropping one bead in each pit until her hand is empty. If it is odd, she moves her hand anti-clockwise dropping one bead in each pit until her hand is empty. 3: Then she looks at the next pit.

- If it has beads, she picks them up and continues from step 2.
- If it is empty the game ends.

For example:



If she picks up two beads from Pit B, she will move clockwise and drop one bead in Pit C, one bead in Pit D and stop. She will then look at pit A.

Chikki starts by picking beads up from Pit A. When the game ends, which pit has the most beads?

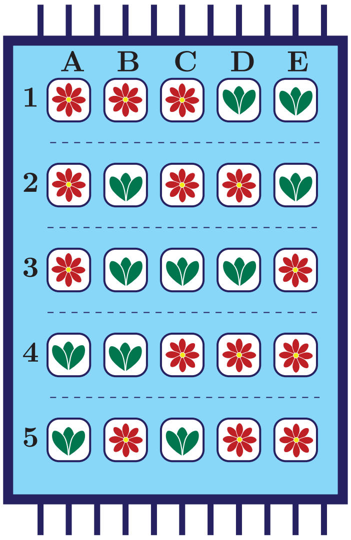
A) Pit A
B) Pit B
C) Pit C
D) Pit D

18 The Carpet Weaver's Mistake

Ali is learning to design Persian carpets. His teacher, Master Weaver, taught him two types of knots: Flowers knot and Leaves knot . Ali is trained by his Master by making a small 5-by−5 carpet. To obtain an aesthetic carpet, Master Weaver set two strict rules:

- Leaf Rule: Every Row must have an even number of Leaf knots (0, 2, or 4).
- Flower Rule: Every Column must have an odd number of Flower knots (1, 3, or 5).

Ali has finished his small carpet, but he accidentally used the wrong knot in exactly one cell.



Which knot did Ali get wrong?

A) Row 2, Column C

B) Row 3, Column B

C) Row 3, Column C

D) Row 4, Column B

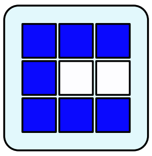
19 Magic Spy Sheets

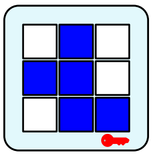
Agent Fox needs to deliver a Secret Map to his base. To hide the information from spies, he uses a system with two clear plastic sheets:

- A **Key Sheet**, which is kept safely at the base and
- A **Message Sheet**, which is the one he will actually send.

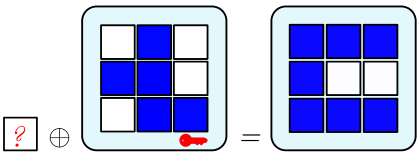
When the base receives the Message Sheet, they stack it on top of the Key Sheet. Then the squares appear magically like this:

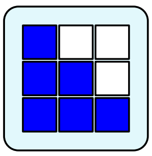
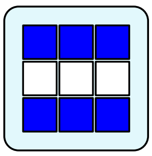
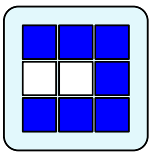
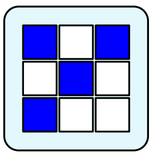
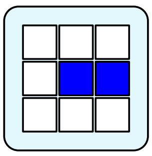
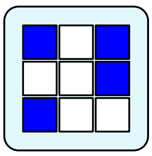
- Blue $\oplus$ Clear = Blue
- Clear $\oplus$ Blue = Blue
- Clear $\oplus$ Clear = Clear
- Blue $\oplus$ Blue = Clear (Magic! The color disappears!)

Agent Fox needs to deliver this Secret Map: 

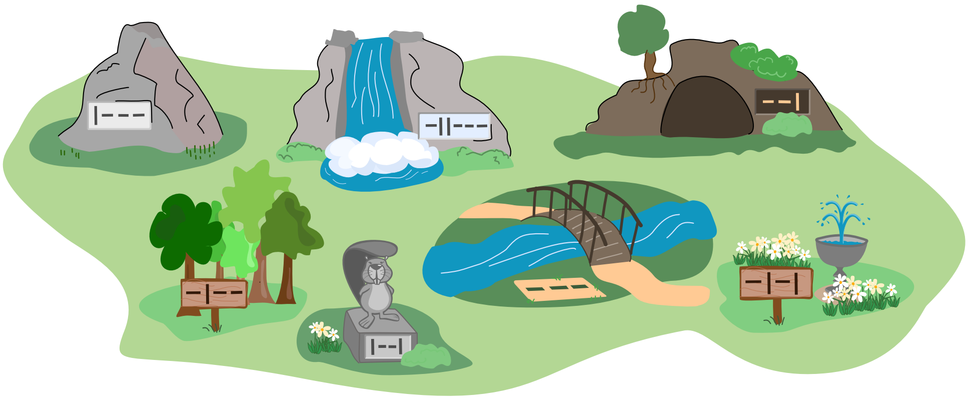
And the base will use this Key Sheet: 

Which Message Sheet should Agent Fox send?



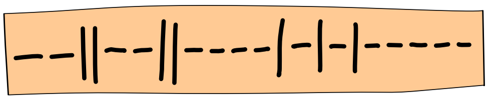
A.  B.  C.  D.  E.  F. 

20 Ben's Walk



Ben took a walk through the park and visited several sites.

Each site has an ID made from the two symbols and |. As he walked, Ben recorded the IDs on the sheet below, from left to right, in the order he passed the sites.



Ben did not pass any site more than once.

Which sequence of sites did Ben visit?

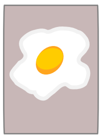
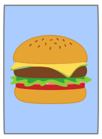
A) Cave → Rock → Beaver Statue → Fountain → Trees → Bridge
B) Cave → Beaver Statue → Rock → Fountain → Trees → Bridge
C) Cave → Beaver Statue → Fountain → Trees → Rock → Bridge
D) Cave → Beaver Statue → Rock → Trees → Fountain → Bridge

21 Video Recommendation

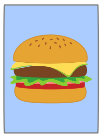
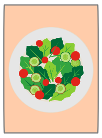
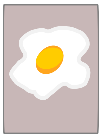
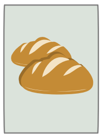
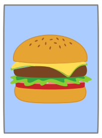
A video app uses a smart rule to suggest new videos:

If another person watched at least 2 of the same videos as you, the app suggests their other videos to you (if you have not seen them).

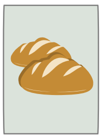
You have recently watched these three videos:

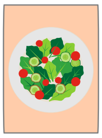
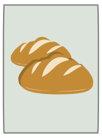
		
How to cook eggs	Italian Pasta Recipe	Burger Recipe

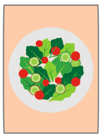
Here is what three other people watched:

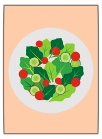
Anu	 Italian Pasta Recipe	 Burger Recipe	 Salad Recipe
Bela	 How to cook eggs	 Baking Bread	 Burger Recipe
Charu	 Italian Pasta Recipe	 Baking Bread	 Street Food

According to the rule, which new videos will the app recommend to you?

A)  

B)  

C)   

D)  

22 Beaver Dam Repair

After a storm, part of the beaver dam has collapsed. Five beavers Asha, Bimal, Chitra, Dev, and Esha rush to carry out repairs.

Each beaver performs exactly one task, and each task is performed by exactly one beaver. The repair tasks are:

- Stabilizing logs
- Redirecting mud
- Cutting wood
- Fixing joints
- Clearing debris

Each beaver knows how to do two of the tasks, as shown in the table below.

Beaver	**Tasks they can perform**
Asha	Stabilizing logs, Redirecting mud
Bimal	Stabilizing logs, Cutting wood
Chitra	Clearing debris, Redirecting mud
Dev	Fixing joints, Cutting wood
Esha	Clearing debris, Fixing joints

Which of the following assignments is not possible?

A) Chitra clears debris and Dev cuts wood

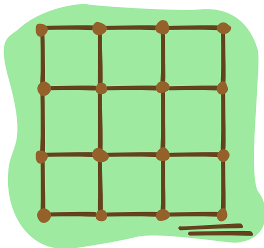
B) Esha clears debris and Bimal cuts wood

C) Asha stabilizes logs and Dev fixes joints

D) Esha clears debris and Bimal stabilizes logs

23 Muddy or Clear?

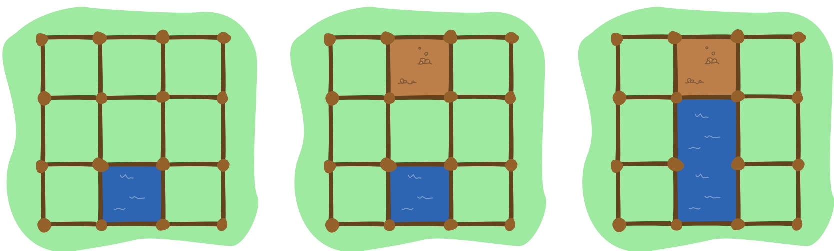
The beavers play a game with clear and muddy water. First they build a grid like this using posts and removable walls:



After, they start playing, they always follow these rules:

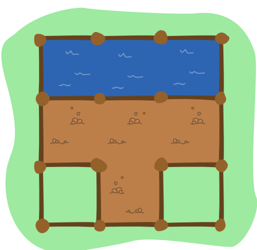
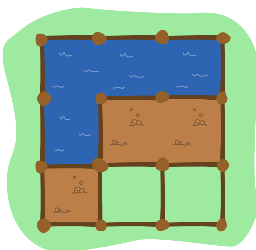
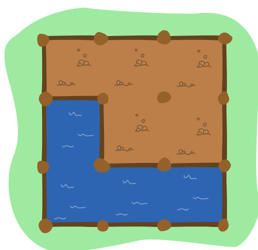
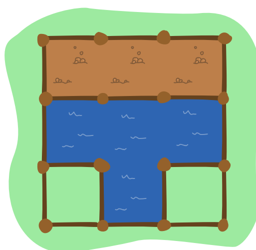
- Luca pours clear water into one randomly chosen field.
- Mora pours muddy water in any other field.
- Luca and Mora then take turns removing one wall at a time. Luca starts first. When a wall is removed, the players water flows into a neighboring field.
- A wall may not be removed if:
  - it is an outer wall of the playing area;
  - there is no water on both sides;
  - it has muddy water on one side and clear water on the other side.
  - The player who cannot remove the wall loses the game.

Here an example of first 3 turns:



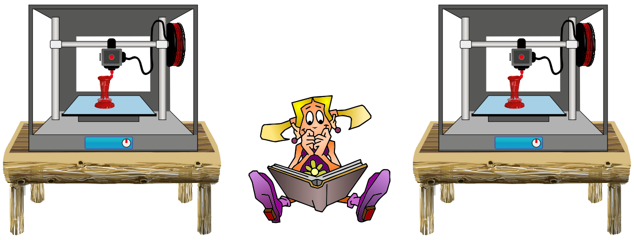
Luca Mora Luca Mora

Only one of these states is possible in the game. Which one?

A)  B)  C)  D) 

24 3D Printing

A town library has two 3D printers that are available for people to use when they are in the library. If a person arrives and a printer is available, he or she starts their print job immediately. Otherwise that person must wait until a printer becomes available. If more than one person is waiting, when a printer becomes available, the next person is chosen randomly by the librarian.



One day 6 people wanted to use the 3D printers. Their arrival time and the time it will take to complete their print job is shown in the table.

Person	Arrival Time	Length of Print Job
Abby	8:00 am	5 hours
Bojan	10:00 am	3 hours
Chloe	12:00 pm	2 hours
Dipika	1:00 pm	4 hours
Edi	2:00 pm	1 hour
Foster	2:00 pm	6 hours

What is the earliest possible time that the last print job will finish?

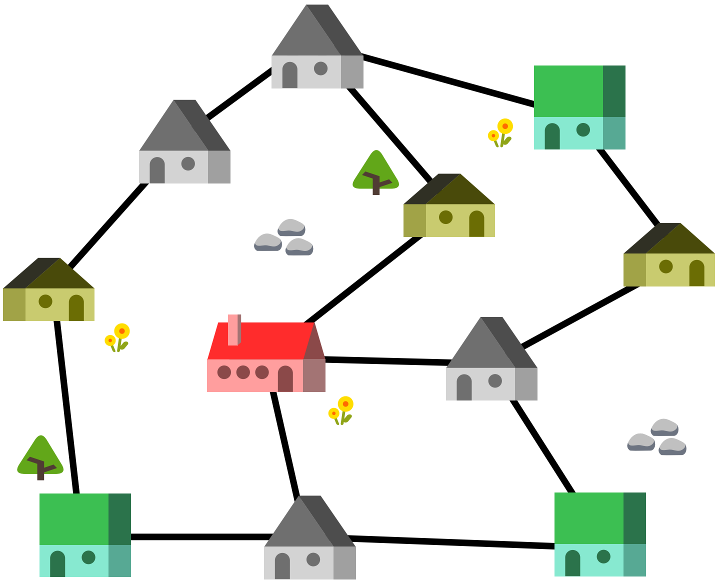
A) 8:00 pm

B) 9:00 pm

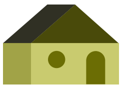
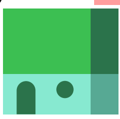
C) 10:00 pm

D) 11:00 pm

25 Gifts for Grandchildren



The picture above shows a map of the houses and roads in the village. Each road (black line) between two houses is exactly 1 km long.

Grandma lives in the big (red) house , and her grandchildren in the smaller (yellow) houses  . The green houses  are toy stores and the grey houses ![greyh] are where other people live.

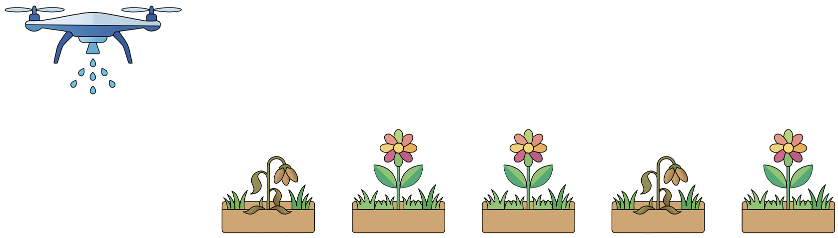
Grandma wants to buy gifts at (exactly) one of the toy stores and deliver them to her grandchildren in all three yellow houses.

What is the minimum distance Grandma needs to travel (in km) that takes her from her house to one of the shops, then to each of the three houses of her grandchildren (to deliver the toys) and then back again to her house?

A) 9 km
B) 10 km
C) 11 km
D) 12 km

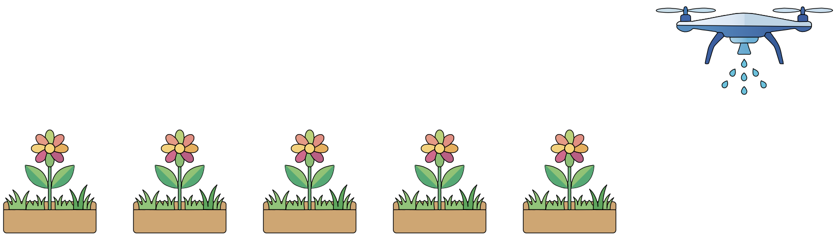
26 Watering Drone

Dr. Beaver tried to develop a drone that automatically waters wilted flowers to make them bloom. She is not entirely sure that it works perfectly.



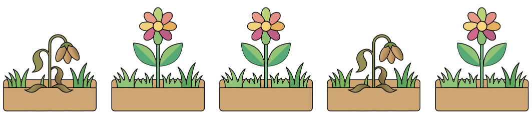
Dr. Beaver programmed the drone as follows. When the drone starts above the leftmost flower box, it repeats the following instructions in order until it stops:

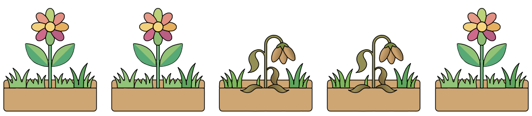
- Check the box directly below. If there is a wilted flower in it, water it. Otherwise, move to the box on the right.
- If the drone is now no longer over a flower box, stop.
- Move to the flower box on the right.

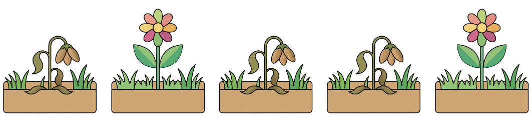


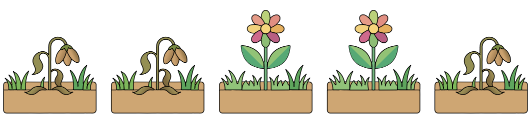
Dr. Beaver suspects that the drone sometimes fails to water all the flowers.

Which of the flower beds (A-D) will not be watered correctly?

A) 

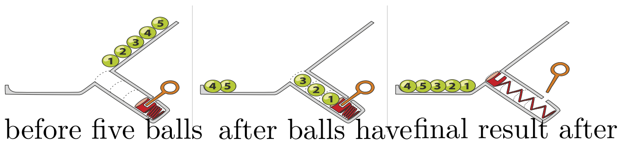
B) 

C) 

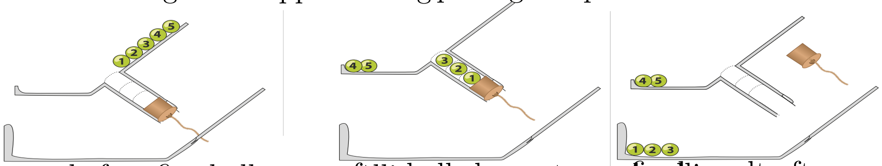
D) 

27 Slope

Numbered balls roll down ramps. When a ball comes to a hole, if the hole is not full, the ball falls in. Otherwise, the ball rolls past the hole. There are 2 kinds of hole stoppers. A stopper-pin  at the side of each hole can be pulled which ejects the balls. The cork  at the bottom can be pulled and just pass the ball to the tray below. Here is an example:

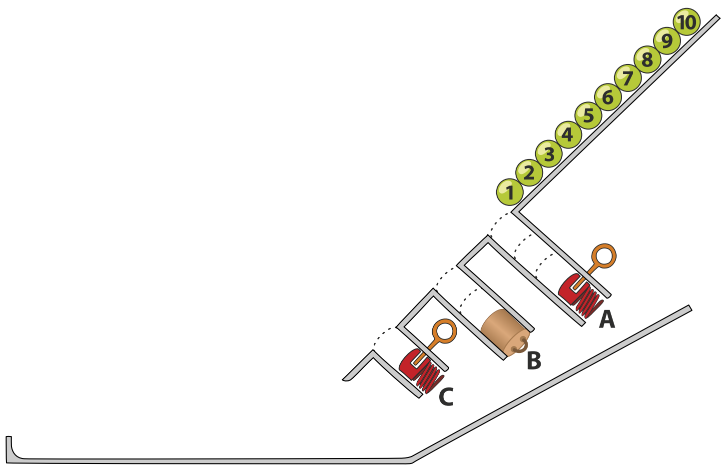


before five balls start rolling | after balls have stopped rolling | final result after pulling the pin



before five balls start rolling | after balls have stopped rolling | final result after removing the cork

Ten balls roll down the ramp below. Three holes A, B and C have space for 3, 2 and 1 balls as shown. The holes A and C have a stopper-pin at the end and the hole B has a cork. Pins and/or corks are pulled in the order A, B, C but each time, only after all balls have stopped rolling.



Which option shows the final result?

A) 

B) 

B) 

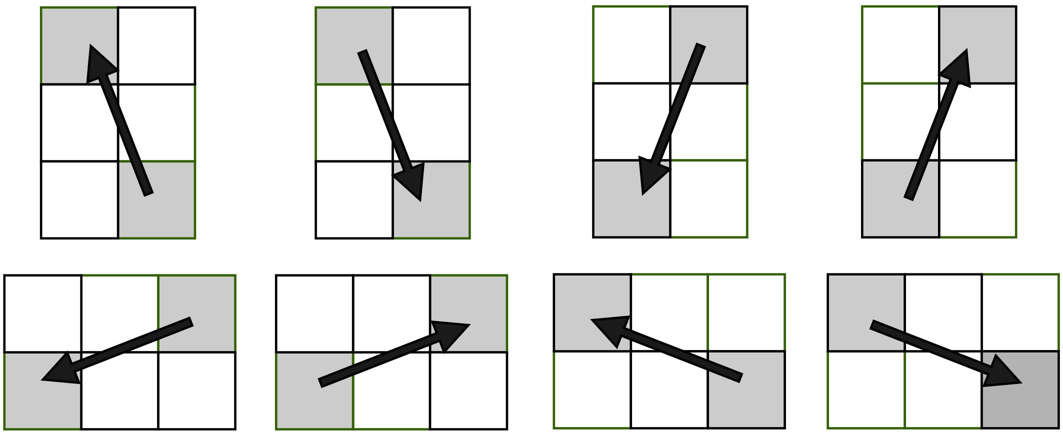
D) 

28 Jumping Horse

A horse is jumping through a field that is split up into 25 squares.

There are two rules:

- It can only visit each square once.
- It can only jump in the following manner:



The horse starts at the square marked 21 and wants to visit every single numbered square. Jumps into white squares are not allowed.

1		3		
	7			10
	12	13	14	
16		18		
21				25

In what order will the horse jump through the field?

A) START 21, 12, 3, 10, 13, 16, 7, 18, 25, 14, 1 FINISH
B) START 21, 12, 3, 14, 25, 7, 16, 13, 10, 18, 1 FINISH
C) START 21, 18, 25, 14, 7, 16, 13, 10, 3, 12, 1 FINISH
D) START 21, 18, 7, 16, 13, 10, 3, 12, 1, 25, 14 FINISH

29 Beaver Juice Shop

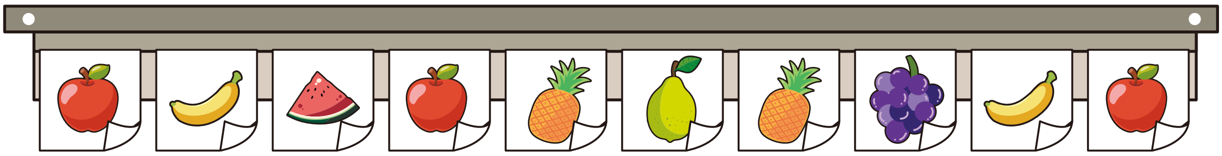
There are exactly three blenders in a beavers juice shop.

The beaver can operate only one blender at a time and wants to avoid mixing flavors.

At the start of the day all three blenders are clean. Then the beaver satisfies requests following these rules:

- If a used blender matches the current request flavor: use it.
- Otherwise, if a clean blender is available: use it.
- If no matching or clean blender exists: clean the blender that has not been used for the longest time and use it.

Todays juice flavor requests in order (from left to right) are:



How many total cleanings are needed to complete all these requests?

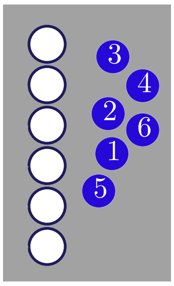
A) 4

B) 5

C) 6

D) 9

30 Coins



There are 6 coins, labeled with the numbers 1 to 6. There are also 6 slots, which are likewise labeled 1 to 6.

A. At the beginning, the coins are placed randomly in the slots. The following procedure is then carried out to place all coins into their correct slots.

B. Consider the slots from top to bottom and take the first coin that is in the wrong slot. Place this coin to the right of the slot whose label matches the value of the coin.

This empties the slot from which the coin was taken.

C. Into this empty slot, place the coin whose number corresponds to that slot.

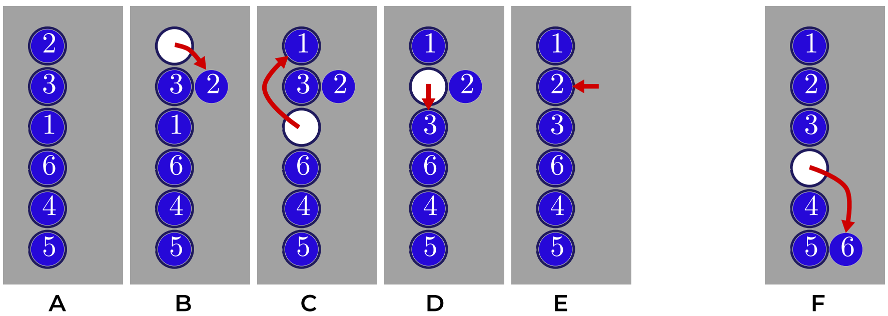
D. This makes another slot empty, which is then filled with the coin that belongs there.

This process continues until the slot next to the coin that was placed aside becomes empty.

E. The coin at the side is then placed into this slot, so that all slots are occupied again.

The entire sequence of these movements (A-E) is called one turn.

F. The procedure can then be repeated with another turn until all coins are in their correct slots

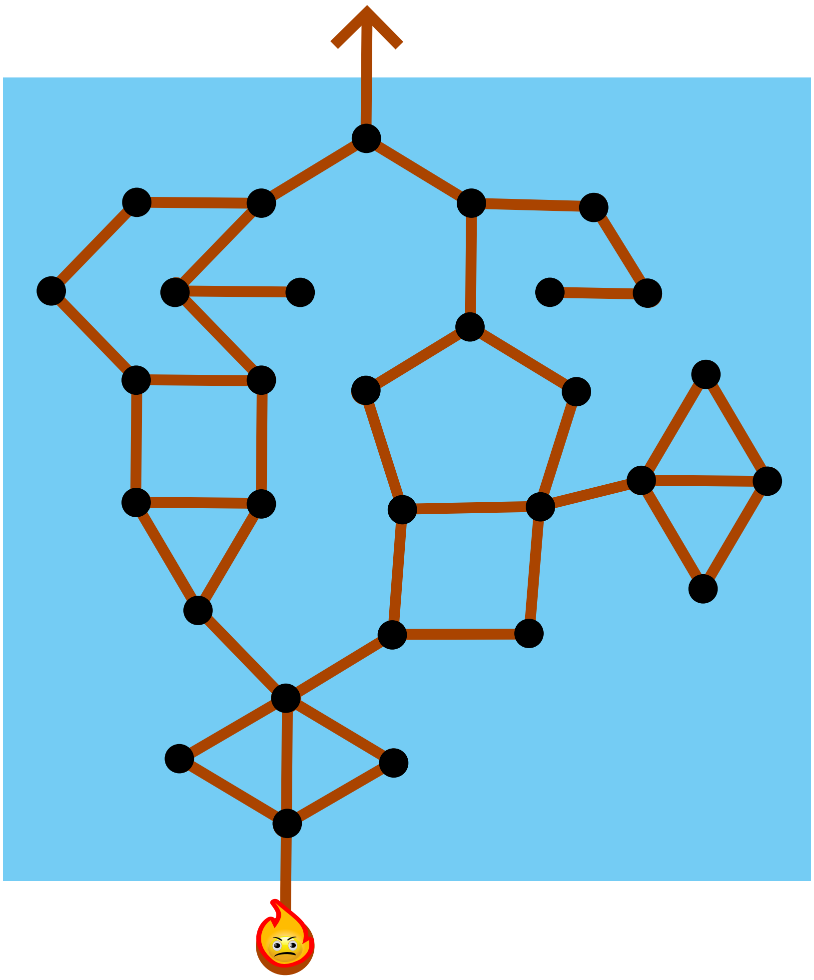


Assuming that initially no coin is in the correct slot: What is the minimum number of turns required to move every coin to the correct slot? What is the maximum number of turns that may be required for that?

A) 1, 6
B) 2, 3
C) 1, 3
D) 2, 6

31 Fire Sprite

There are a number of islands on the lake, connected by footbridges.



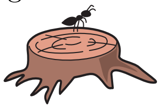
The fire sprite (a fairy tale character) wants to get from where he is standing, up to the arrow. He wants to visit as many islands as possible. He cannot swim because the water would extinguish him.

However, the footbridges are wooden, and will burn down when the fire sprite walks on them. So the fire sprite cannot use any footbridge twice.

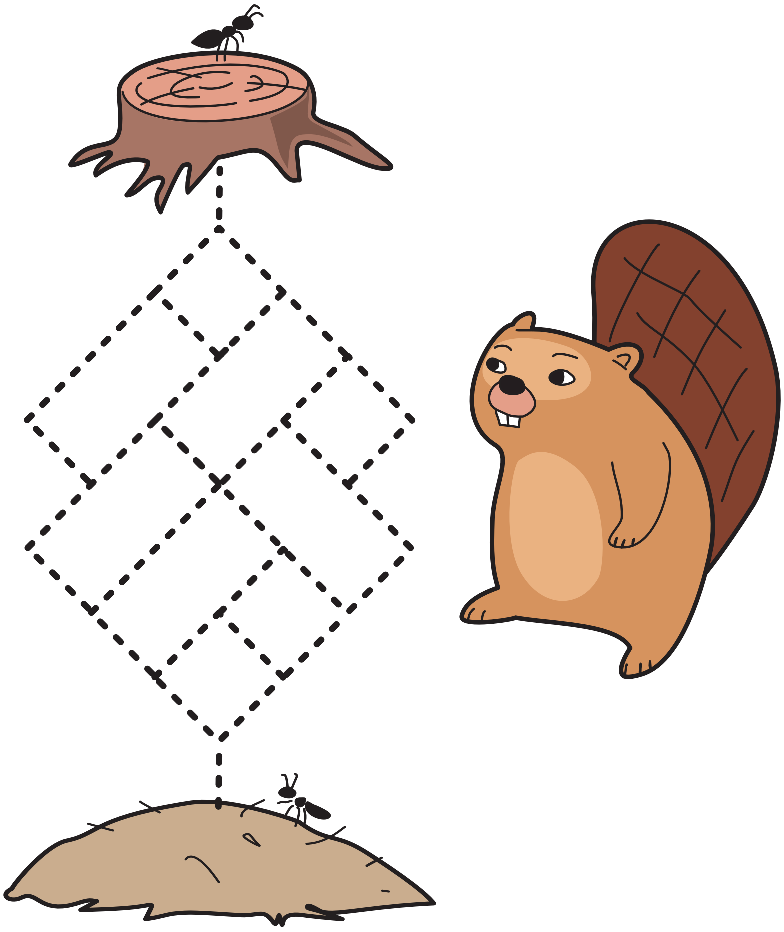
How many islands can the fire sprite visit at most?

A) 12
B) 13
C) 14
D) 15
E) 20

32 The Anthill Puzzle

One day, a beaver was watching ants walking around and wondered how many different paths the ants take from the tree stump  to the anthill .

The beaver drew a simple map, marking all possible paths with a dashed line. The ants want to get to the anthill as quickly as possible, so they never go back towards the tree stump.

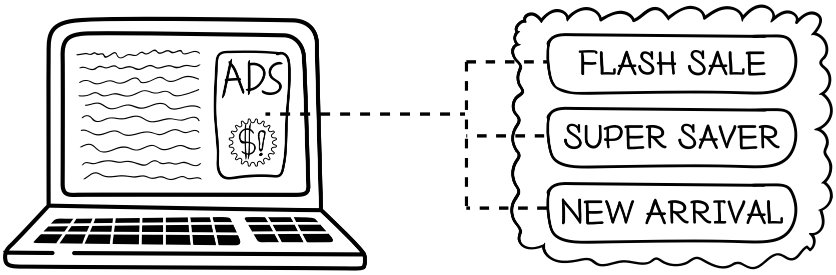


How many different paths take the ants from the tree stump to the anthill?

A) 15
B) 16
C) 17
D) 18

33 Online Ad

An online shop shows one advertisement each day on its webpage. There are three ads: Flash Sale, Super Saver and New Arrival. Each day the shop earns beaver coins from the ad shown. Over time, the online shop would like to earn as many beaver coins as possible.



The shop keeps a score for each ad. The score shows how good the ad has been so far.

The shop follows these rules to decide which ad to show:

- At the beginning all scores are zero.
- For the first three days, the shop shows one different ad each day, so that every ad is shown once.
- After an ad is shown on a day, its score is updated like this:

The new score of the ad is the average of its previous score and todays earnings. The scores for the other ads do not change.

- On every 3rd day after that (Days 6, 9, 12,) the shop chooses the ad with the lowest score to explore alternatives.
- On all other days, the shop shows the ad with the highest score.
- If two or more ads have the same score, the shop chooses Flash Sale over Super Saver over New Arrival, in that order.

So far

- Day 1: earning = 20 beaver coins
- Day 2: earning = 20 beaver coins
- Day 3: earning = 80 beaver coins
- Day 4: earning = 20 beaver coins
- Day 5: earning = 20 beaver coins

Which ad will the system show on day 6?

A) Flash Sale
B) Super Saver
C) New Arrival
D) Cannot be decided

34 Beaver Tiles

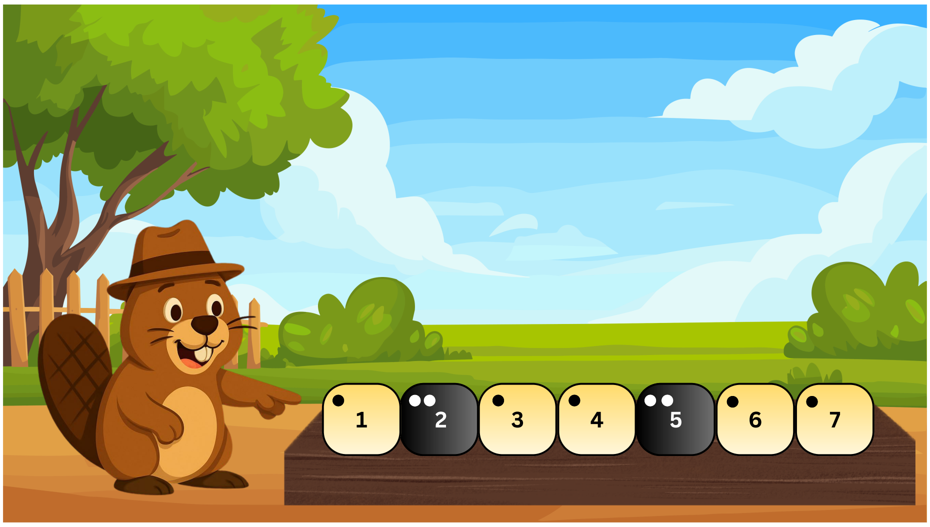
Beaver Kito has a row of 7 tiles. Each tile is Light on one side and Dark  on the other side.

In each move, Kito chooses exactly two tiles and flips both of them, so that:

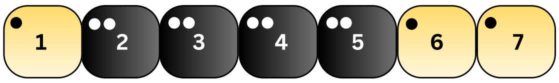
- Light becomes Dark
- Dark becomes Light

Kito can repeat this move as many times as he wants.

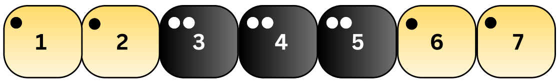
The starting pattern of the tiles is shown in the picture.



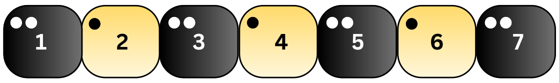
Which of the following patterns is impossible to reach from the starting pattern?



A)



B)



C)

D)

35 Tile Factory

A tile factory has two tile printing machines ( and ). To make a decorative tile, choose a stencil pattern and a tile and put both in one of the machines. The machine will then print the pattern on the tile. The two machines work a bit differently.

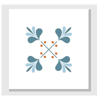
Pattern: 

Machine: 

Input Tile: 

How it Works: The machine repeats the following steps 4 times:

- print a stencil pattern on the top left quarter of the tile,
- then rotate the tile 90ř clockwise.

Output Tile: 

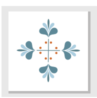
Pattern: 

Machine: 

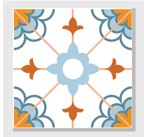
Input Tile: 

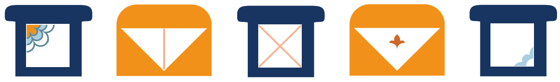
How it Works: The machine repeats the following steps 4 times:

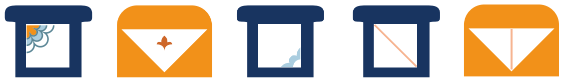
- print a stencil pattern on the top triangle of the tile,
- then rotate the tile 90ř clockwise.

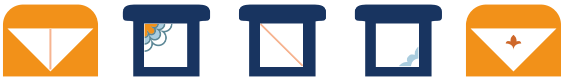
Output Tile: 

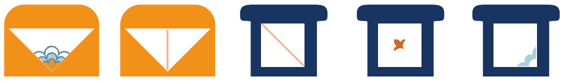
A tile can be used as input tile more than once. When it is printed on more than once, a pattern printed later will be printed over all earlier patterns.

Which of the following sequences of machines and stencil patterns will produce this  *tile?*

A) 

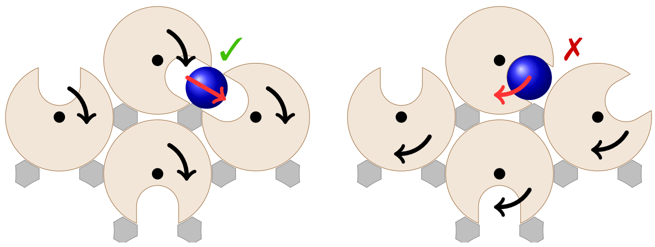
B) 

C) 

D) 

36 Motorized Rollers

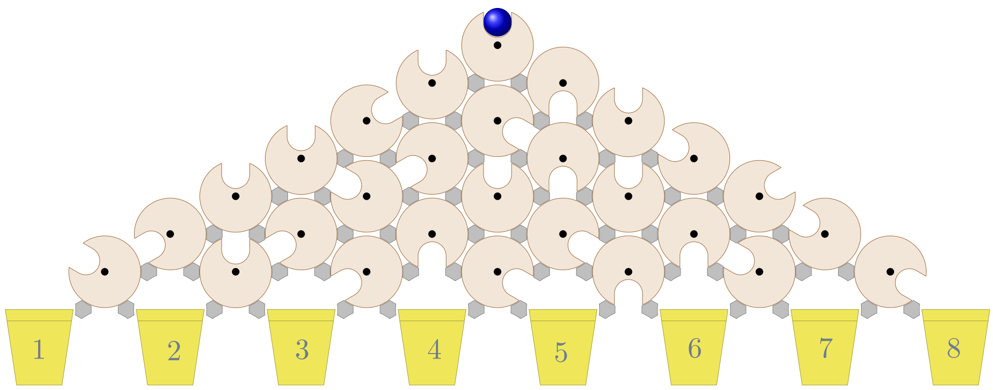
A ball is sent through an array of identical motorised rollers. The rollers slowly rotate in the same direction at the same steady rate. There is a switch that determines whether the rollers are all simultaneously rotating either clockwise or anticlockwise.



Each roller has a slot into which a ball can fit perfectly. When the slots of adjacent rollers line up with each other, the ball will drop from the higher roller into the lower one. (As in the left picture.) In the configuration on the right, the gaps will never align (given that the rollers all move in the same direction) and the ball will remain in the top roller.

As shown in the picture, plugs are placed between the rollers to ensure the ball can never drop into the gap between rollers or out of the array of the rollers.

A set of rollers is set up as below, with buckets placed to catch the ball.



How many of the buckets could the ball eventually reach?

A) 1

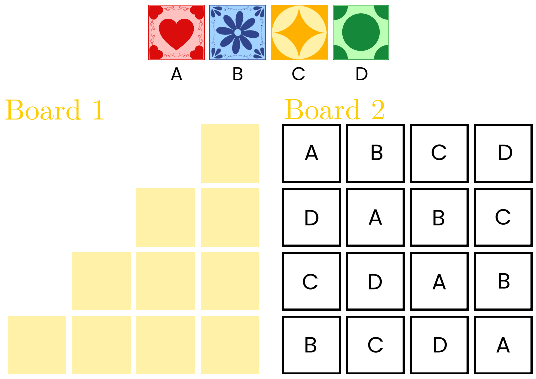
B) 2

C) 3

D) 4

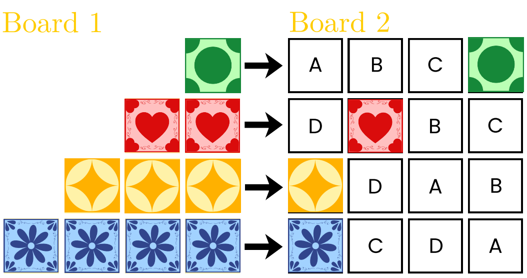
37 Azul Board Game

The beaver Laís is playing a board game. She has tiles of four different types (A, B, C, and D) and two boards (1 and 2). The letters on Board 2 indicate the type of tile that can be placed in each position.



The game consists of several rounds. In each round, Laís starts by placing tiles on Board 1. She can place as many tiles as she wants on Board 1 as long as each row contains the same type of tile, and each row is either complete or contains no tiles.

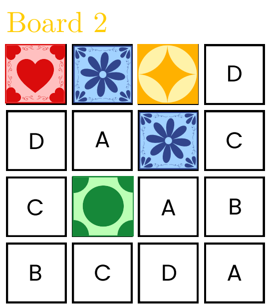
Then, for each completed row on Board 1, Laís takes one tile from that row and places it in the same row on Board 2, in the column corresponding to its type. The remaining tiles from that row on Board 1 are discarded.



This shows a possible result after the first round of the game.

The goal of the game is to complete the largest possible number of rows and columns on Board 2.

After few rounds of the game, Board 2 was:



In the next round, Laís wants to use at most 8 tiles. Which type of tile should she not use if she wants to complete the largest possible number of rows and columns on Board 2?

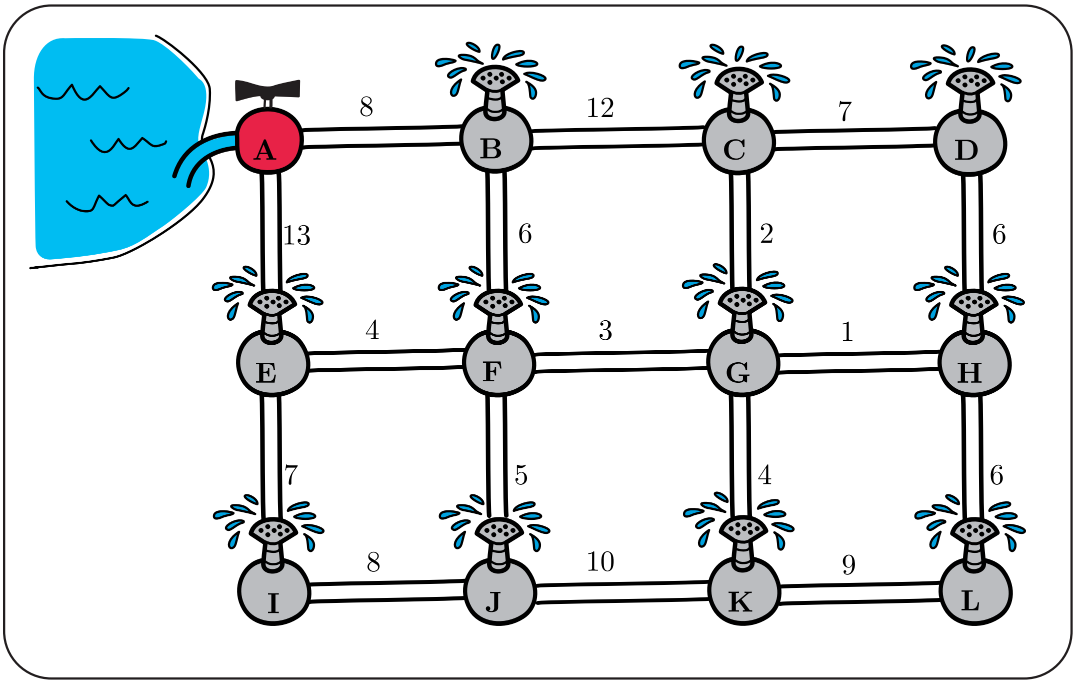
A)  B)  C)  D) 

38 The Best Pipe Plan

Beaver Ben is installing a watering system in his field. Water starts from the main source A and must reach all sprinklers in the garden. Each sprinkler can be connected to at most three pipes.

Each pipe has a number indicating the cost of building it.

Ben wants to build the system as cheaply as possible, while still ensuring that water reaches every sprinkler.



What is the smallest total cost required to build such a network?

A) 57

B) 59

C) 60

D) 63

39 Free Taco Order

Tacos are a unique part of Mexican culture: varied, delicious, and prepared fast!

A Taquería (food truck) promises:

ńIf your group order takes more than 15 minutes, the meal is FREE!ż

This applies to orders of up to 8 tacos (any combination you want).

You and your friends want a free meal. The waiters assign the tacos upfront to the three cooks Ana, Beto and Carlos one by one, in the exact sequence in your order. They do it following these rules:

- Rule 1 (Fastest Cook): Give the new taco to the cook who prepares it the fastest.
- Rule 2 (Overload Protection): If that cook already has 10 minutes or more of work waiting, give the taco to next fastest cook instead.
- Rule 3 (Tie-Breaker): If cooks tie in speed, give it to the cook with the least amount of work. If there is still a tie, assign it in the next order: Ana, Beto, Carlos.
- Rule 4 (System Overload): If ALL cooks are overloaded (>= 10 mins), ignore Rule 2 and give the taco to the fastest cook.

Each cook has different speeds for each taco type.

See the table below of Preparation times in minutes:

Taco	Ana	Beto	Carlos
Paneer	3	5	5
Chicken	5	3	5
Veggie	2	11	11

Cooks prepare one taco at a time in the assigned order.

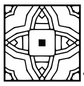
The total time stops when the last taco is finished.

Which of the following order sequences will successfully trick the Taquería rules and take more than 15 minutes?

A) 8 Veggie Tacos.
B) 8 Paneer Tacos.
C) 4 Paneer, 2 Chicken, and 2 Veggie Tacos.
D) 4 Chicken and 4 Veggie Tacos.
E) It is not possible

40 Painting The Floor

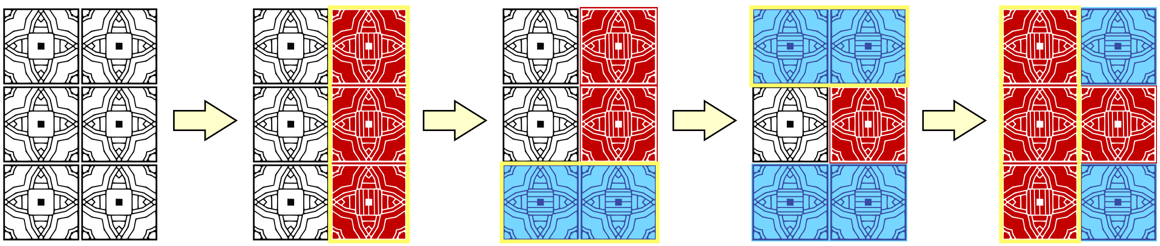
In order to prepare the room for the Bebras Convention, the organizers decided to paint the floor in blue and red.

The floor can be seen as a grid of initially colorless square tiles . The organizers can only do two types of operations:

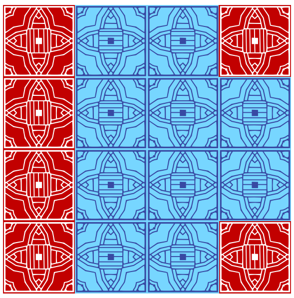
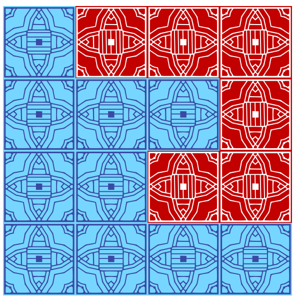
-  paint an entire row in blue
-  paint an entire column in red

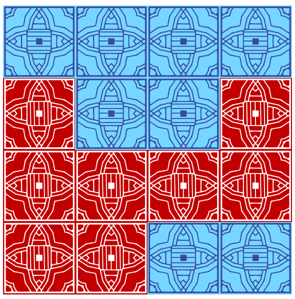
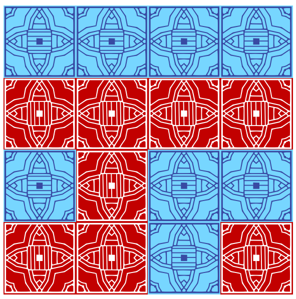
Tiles can be painted multiple times and their color will only depend on the last ink used.

For example, if the floor was a 3 rows by 2 columns grid, the following sequence of operations could be made:



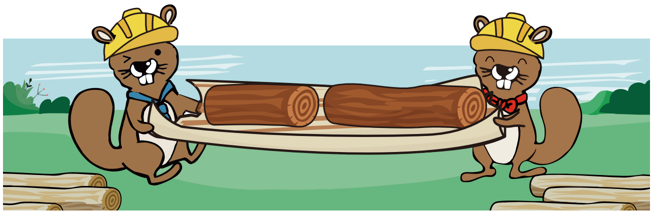
Which of the following 4×4 colored floors cannot be obtained starting from an initially colorless grid?

A.  B. 

C.  D. 

41 Beaver Dam

Beaver engineers are building a dam and need a log that is 21 meters long.



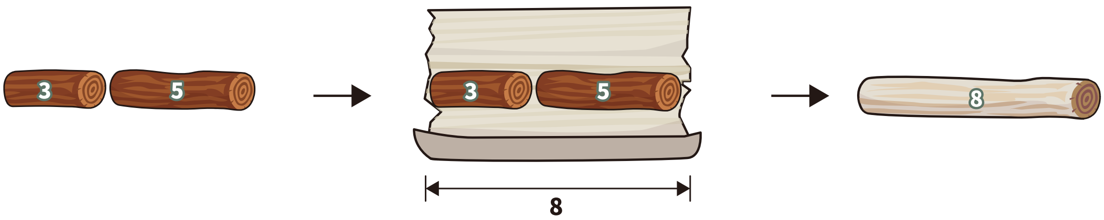
However, they only find six shorter logs which are 1, 2, 3, 4, 5, and 6 meters. They must join these six logs to form one log that is exactly 21 meters.

To keep the log strong, they follow these rules:

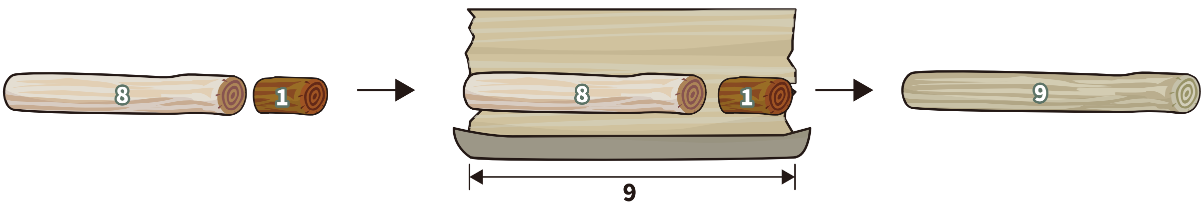
Each round, join exactly two logs end to end into one new log.

- Tightly wrap the entire new log with tree bark. The bark needed equals the length of the new log.
- The new log is treated as one log and can be joined with another log later.
- Take three logs with length 1, 3, and 5 meters for example:

If they join the 3-meter and 5-meter logs first, they need 8 meters of bark.



If they then join this 8-meter log with the 1-meter log, they need another 9 meters of bark.



When they join all six logs into one 21-meter log, what is the minimum total length of bark needed?

A) 45

B) 51

C) 56

D) 63

A Authors

James Atlas

Masiar Babazadeh

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Maria Cepeda

Valentina Dagien

Christian Datzko

Asterios Karagiannis

David Khachatryan

Vaidotas Kinius

Lukas Lehner

Yoshiaki Matsuzawa

Hamed Mohebbi

Mattia Monga

Jean-Philippe Pellet

Zsuzsa Pluhár

Wolfgang Pohl

Pedro Ribeiro

Dirk Schmerenbeck

Vipul Shah

Jacqueline Staub

Alieke Stijf

Susanne Thut

Christine Vender

Michael Weigend

Chris Wetherell

Kyra Willekes

Seth Ayim

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Emily Cavalcanti

Haris Gavranovic

Marvin G. Hall

Heidi Kaarto

Viadotas Kincius

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B Partners

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Raspberry Pi Foundation
https://https://www.raspberrypi.org/

Tata Consultancy Services Foundation
https://https://www.tcs.com/who-we-are/corporate-social-responsibility/

Tata Steel Foundation
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